

Electrons in Crystal

In Hydrogen Atom



It has the Hamiltonian of

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(r)$$

$$= -\frac{\hbar^2}{2m} \nabla^2 + V(r)$$
 with ground state ψ_{100}

With two atoms, one electron can be in a superposition



The superposition phase for this one electron creates different energy levels



At certain a , degeneracy is lifted at N
 But at $a \rightarrow \infty$, the energy is degenerate again.

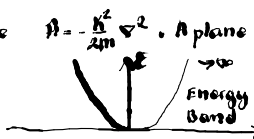


Even though, as $a \rightarrow \infty$ it seems like the electron should be localized for only one atom. That is correct except that think of it as you start with a superposition and separate or you start with a "single" atom. This thinking gives

localized or "localized"

Consider this electron as free particle is the eigenstate

$$\psi \rightarrow e^{i\vec{k} \cdot \vec{r}}$$



Note that as $\vec{k} \cdot \vec{r}$, it does not repeat. Lattice allows the repetition

But in a lattice, there is also a potential:

$$\left(-\frac{\hbar^2 \nabla^2}{2m} + V(\vec{r}) \right) \psi(\vec{r}) = E \psi(\vec{r}) \quad \text{with } V(\vec{r} + \vec{R}_n) = V(\vec{r})$$

The solution becomes:

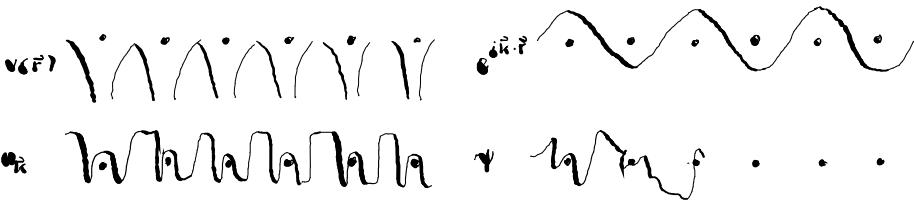
$$\psi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}(\vec{r})$$

$u_{\vec{k}}(\vec{r} + \vec{R}_n) = u_{\vec{k}}(\vec{r})$
 make $\vec{k}$ discrete
 v : band index / energy level

The Bloch theorem says

$$\psi_{\vec{k}}(\vec{r} + \vec{R}) = \psi_{\vec{k}}(\vec{r}) e^{i\vec{k} \cdot \vec{R}}$$

where $\vec{k} \cdot \vec{R}$ can be anything!



Thus for a general structure.

$$\left(-\frac{\hbar^2 \nabla^2}{2m} + V(\vec{r}) \right) \psi_{\vec{k}}(\vec{r}) = E_{\vec{k}} \psi_{\vec{k}}(\vec{r})$$

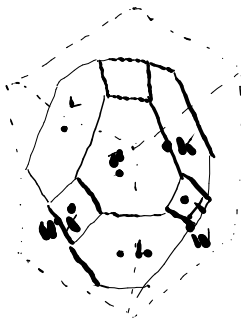
V : lattice potential $E_{\vec{k}}$: energy band structure

$$\psi_{\vec{k}+\vec{G}}(\vec{r}) = \psi_{\vec{k}}(\vec{r}) \quad E_{\vec{k}+\vec{G}} = E_{\vec{k}}$$

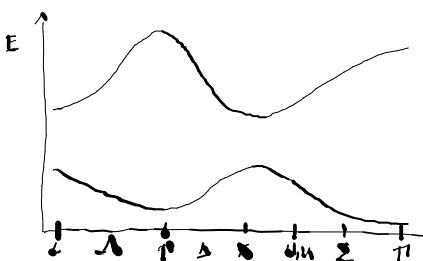
Even when we look at $\vec{R}_n$ away from last position of the electron wave vector is increased by $\vec{G}$. The energy is the same.

Observe the reciprocal lattice of an FCC crystal (BCC reciprocal), We start with $\vec{r} = \vec{k} = 0$

- X is center of square face
- L is center of hexagon face
- W is vertex
- K midpoint of edge between two hexagon
- M midpoint of a square and hexagon
- A: $\Gamma \rightarrow X$
- D: $\Gamma \rightarrow L$
- G: $L \rightarrow W$
- Σ : $X \rightarrow U$



When a band structure plot has a horizontal axis such as $\Gamma \rightarrow X \rightarrow W \rightarrow K \rightarrow L$, it means we calculate the energy along these directions. k in the x-axis is not magnitude



Comments: The Bloch theorem allows $\vec{k} \cdot \vec{R}$ to be any phase. This says that any electron with any $\vec{k}$ is allowed in the crystal. However a $\vec{k} + \vec{G}$ is reproducing the same wave function. $\vec{k}$ is discrete: allowed shift while $\vec{R}$ is continuous in an infinite crystal.

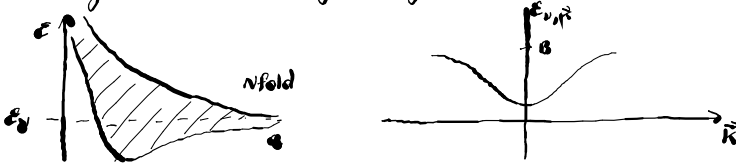
Tight Binding Model

Consider putting atoms closer such as there is an overlapping with wave functions



The overlapping lifts up N -fold degeneracy

this creates a continuous band



We will see how this energy band arrives, Consider the Schrödinger

$$-\frac{\hbar^2}{2m} \nabla^2 + W(\vec{r}) \psi_{\vec{k}}(\vec{r}) = E_{\vec{k}} \psi_{\vec{k}}(\vec{r})$$

In one dimension model $\vec{r} \rightarrow x$

$$W(x) = \sum_{i=1}^N W_{i0}(x-R_i) \quad \text{with } R_i = i \cdot a \quad N \rightarrow \infty$$

The solution for each atom is

$$\psi_0 \psi_i(x) = E_0^0 \psi_i(x) \quad \text{with } H^0 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + W_{i0}(x)$$

Construct

$$\psi_{\vec{k}}(x) = \frac{1}{N} \sum_i e^{ikR_i} \psi_0(x-R_i) \quad E_{\vec{k}} = \frac{\int dx \psi_{\vec{k}}^* H \psi_{\vec{k}}}{\int dx \psi_{\vec{k}}^* \psi_{\vec{k}}}$$

This is a superposition of an electron with a phase e^{ikR_i} . This phase changes from one site to other.

$$E_{\vec{k}} = \frac{1}{N} \sum_{nm} e^{-ik(R_n - R_m)} \int dx \psi_0^*(x-R_n) \{E_0^0 + \Delta V_{nm}(x)\} \psi_0(x-R_m) \quad \Delta V_{nm} = \sum_{l \neq m} W_{l0}(x-R_l)$$

Where other nearby potentials near m contribute the perturbation

Assume small overlap

First order
Second order

$$\int dx \psi_0^*(x-R_n) \Delta V_{nm}(x) \psi_0(x-R_m) \approx V_0 \delta_{nm} + V_1 \delta_{n,m+1} + V_1^* \delta_{n,m-1}$$

$$E_{\vec{k}} = E_0^0 + V_0 + V_1 \frac{1}{N} \sum_{nm} \left(e^{-ik(R_{m+1} - R_n)} \delta_{n,m+1} + e^{-ik(R_{m-1} - R_n)} \delta_{n,m-1} \right) \quad R_{m,0} - R_{n,0} = \pm a$$

$$= E_0^0 + V_0 + 2V_1 \cos(ka)$$

This perturbation theory predicts a cosine structure for the band

An electron near the bottom of a conduction band behaves approximately like a free electron, but with an effective mass

At $\vec{r} = \vec{r} = 0$, expand Taylor series

$$E_{\vec{k}} \approx E_{\vec{k}=0} + \frac{1}{2} \frac{\partial^2 E_{\vec{k}}}{\partial k^2} \Big|_{k=0} k^2 = E_{\vec{k}=0} + \frac{\hbar^2 k^2}{2m_0} \quad \text{At minimum } \frac{\partial E}{\partial k} = 0$$

Free energy

$E_{\vec{k}}$ is dispersion of free particle with effective mass m_0

Periodic Boundary Condition

Until now, the volume is essentially infinite. But, in reality the volume $V=L^3$ is finite.



infinite extension
in each direction

We make $\psi(x+L) = \psi(x)$ in all direction. This is mathematically the same as infinite crystal. That is

$$e^{iK(x+L)} = e^{iKx} \Rightarrow K_L = 2\pi$$

Finite crystal makes the wavelength finite.

$$K_L = m \frac{2\pi}{L} \quad \Delta K_L = \frac{2\pi}{L} \quad \Delta^3 K = \Delta K_x \Delta K_y \Delta K_z = \left(\frac{2\pi}{L}\right)^3$$

As $L \rightarrow \infty$, this K is continuous.

Using

$$\sum_{\vec{k}} = \frac{1}{\Delta^3 K} \sum \Delta^3 K \rightarrow \left(\frac{L}{2\pi}\right)^3 \int d^3 K$$

Density of states can be evaluated

$$\sum_{\text{spin}} \frac{2}{V} \sum_{\vec{k}} f(E_{\vec{k}})$$

$L/V \rightarrow \infty$, $\vec{k}$ is continuous,

we ask how many states have energy near E rather than what are all individual $\vec{k}$ states

Example is the density of electrons in a band

$$n_e = \frac{N_e}{V} = 2 \frac{1}{V} \sum_{\vec{k}} f(E_{\vec{k}}) \rightarrow \int dE g(E) f(E)$$

$$g_V(E) = \frac{\text{number of available states } \vec{k} \text{ around } E \text{ in band } V}{(\text{unit volume})(\text{unit energy})}$$

Formally, one can write as how many $\vec{k}$ contain same energy

$$g_V(E) = \frac{2}{V} \sum_{\vec{k}} \delta(E - E_{\vec{k}}) \Rightarrow \int dE g_V(E) f(E) = \frac{2}{V} \sum_{\vec{k}} f(E_{\vec{k}})$$

Since tight binding model gives

$$E_{\vec{k}} = E_{20} + \frac{\hbar^2 k^2}{2m_0} \Rightarrow k = \sqrt{\frac{2m_0(E - E_{20})}{\hbar^2}}$$

the amount of k -states lie in thin spherical cell is

$$dN = 4\pi k^2 dk$$

$$g_V(E) \propto k^2 \frac{dk}{dE}$$

$$\frac{dk}{dk} = \frac{\hbar^2 k}{m_0} \quad \frac{dk}{dE} = \frac{m_0}{\hbar^2 k} \Rightarrow k^2 \frac{dk}{dE} \propto m_0^{3/2} \sqrt{E - E_{20}}$$

$$g_V(E) = \frac{1}{2\pi^2} \left(\frac{2m_0}{\hbar^2} \right)^{3/2} \sqrt{E - E_{20}} \quad \Theta(E - E_{20})$$



Electron Distribution

Consider how likely each electron state in a band can be occupied

$$f_v(\vec{k}) = \langle \hat{a}_{v\vec{k}}^\dagger \hat{a}_{v\vec{k}} \rangle = \frac{1}{e^{\beta(E_{v\vec{k}} - \mu)} + 1} \rightarrow \text{lower energy allows better occupation}$$

$\beta = 1/(k_B T)$

v : band index $\vec{k}$ is crystal vector and $E_{v\vec{k}}$ is the energy of that state. μ is chemical potential

f indicates the average number of particles in state $v, \vec{k}$

The chemical potential is chosen such that

$$N_e = \sum_{\vec{k}} f_v(\vec{k})$$

Particle density

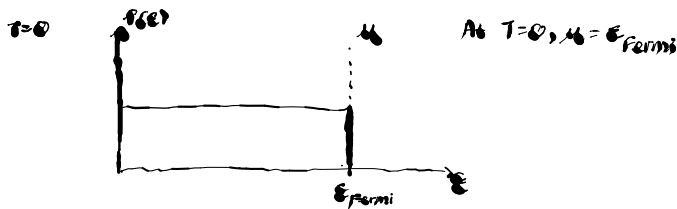
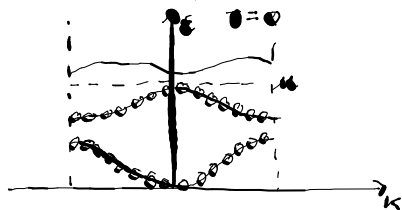
$$n = \frac{N_e}{V}$$

At limit $T \rightarrow 0$, if $E < \mu$, $e^{\beta(E - \mu)} \rightarrow 0$ $f_v(\vec{k}) \rightarrow 1$

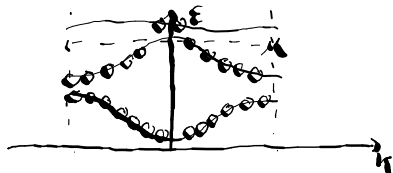
if $E > \mu$, $e^{\beta(E - \mu)} \rightarrow \infty$ $f_v(\vec{k}) \rightarrow 0$

At zero temperature, there is no air particle hitting the electron so that it can only occupy states below μ

When temperature increases, some particles have more energy to occupy above μ



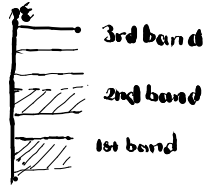
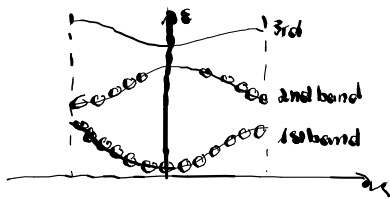
$T > 0$: Fermi edge



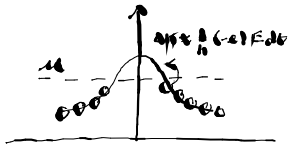
When T is large enough $\beta \rightarrow 0$, this is maxwell distribution $f_v(\vec{k}) \rightarrow e^{-\beta(E_{v\vec{k}} - \mu)}$

Classification of Solid

At Fermi distribution, $k_B T \sim 0.025$ eV, the typical scale for E is eV since $k_B T \sim 0.025$ eV even at 300K

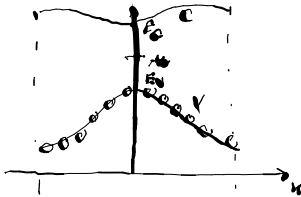


If μ is within a band, it is metal



An applied static E field can easily accelerate electrons in the band producing electrical current

If μ is between bands, it is insulator / semiconductor



The electrons in valence band create no current due to symmetry, but yet has no energy to excite to upper band.

Summary:

